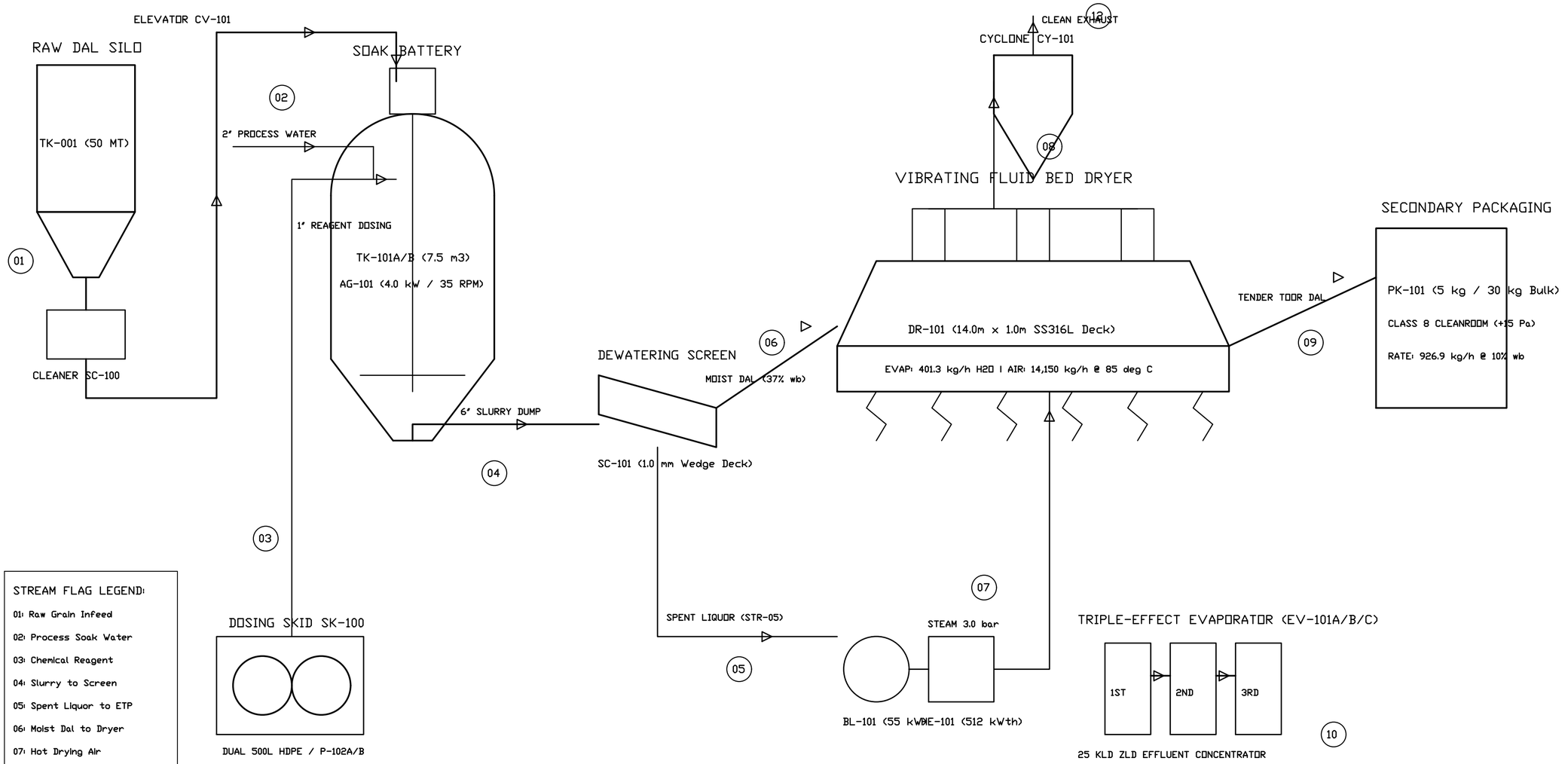
KEY PROCESS DESIGN BASIS & THERMODYNAMIC OPERATING PARAMETERS:

Throughput: 15,000 kg/day as-fed Dal | Operating Schedule: 16.0 h/day (Mean 937.5 kg/h, Design 1,102.9 kg/h @ 85% OEE)

Infeed Moisture: 10.50% w.b. (839.06 kg/h dry solids) | Product Moisture: 10.00% w.b. (926.90 kg/h saleable dal) | Solids Yield: 99.40%

Soak Liquor Ratio: 1.50 : 1.00 (Water:Dal) | Dosing Reagent: Food-Grade NaHCO3 / Citrate | Drying Air: 85 deg C | Deck Area: 14.0 m2

All Product-Contact Metallurgy: AISI SS316L 2B Sanitary Polish (Ra <= 0.8 um) | CIP / Washdown Compatible | ZLD Effluent Evaporation



STREAM FLAG LEGEND:

- 01: Raw Grain Infeed
- 02: Process Soak Water
- 03: Chemical Reagent
- 04: Slurry to Screen
- 05: Spent Liquor to ETP
- 06: Moist Dal to Dryer
- 07: Hot Drying Air
- 09: Saleable Toor Dal

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.		
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project		
PROCESS FLOW DIAGRAM - OVERALL PLANT		
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)		
DWG NO: DWG-FIN-15T-001	REV: V8.0	SHEET: 1 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ

STR	STREAM DESCRIPTION	SOURCE	DESTINATION	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	T (degC)	P (bar g)	PHASE STATE	VOL FLOW (m3/h)
01	Raw Split Pigeon Pea Infeed	Silo TK-001	Elevator CV-101	937.50	839.06	98.44	10.50%	28.0	ATM	Solid Granular	1.17
02	Treated Soft Process Water	Water Plant WTP-01	Soak Battery TK-101	1,406.25	0.00	1,406.25	100.00%	30.0	2.00	Liquid	1.41
03	Reagent Dosing Solution (10%)	Dosing Skid SK-100	Soak Battery TK-101	14.06	1.41	12.65	90.00%	28.0	3.00	Aqueous Soln	0.01
04	Hydrated Dal Slurry Discharge	Soak Battery TK-101	Screen SC-101	2,357.81	840.47	1,517.34	64.35%	30.0	0.20	Solid-Liquid	2.14
05	Separated Spent Soak Liquor	Screen SC-101	Equalization TK-104	1,018.75	6.44	1,012.31	99.37%	30.0	ATM	Turbid Liquid	1.02
06	Moist Dewatered Dal to Dryer	Screen SC-101	Fluid Bed DR-101	1,339.06	834.03	505.03	37.72%	30.0	ATM	Wet Particulate	1.58
07	Hot Conditioned Drying Air	Air Heater HE-101	Plenum DR-101	14,150.00	0.00	155.65	1.10% db	85.0	0.03	Gas	11,800.0
08	Moist Vapour Exhaust Air	Dryer Hood DR-101	Cyclone CY-101	14,551.31	10.60	556.96	3.83% db	62.0	-0.01	Dusty Gas	12,950.0
09	Finished Tender Toor Dal	Dryer DR-101	Packaging PK-101	926.90	834.21	92.69	10.00%	35.0	ATM	Solid Granular	1.16
10	Evaporator Clean Condensate	Condenser HE-103	Water Tank TK-105	980.00	0.00	980.00	100.00%	45.0	ATM	Distillate Liq	0.99
11	ZLD Concentrate Effluent	3rd Effect EV-101C	Solar Pan / Bagging	38.75	15.50	23.25	60.00%	55.0	0.50	Heavy Slurry	0.03
12	Cyclone Separated Fines	Cyclone CY-101	Packaging Recycle	10.60	9.54	1.06	10.00%	58.0	ATM	Fine Powder	0.02

MASS BALANCE RECONCILIATION & SOLIDS CONSERVATION AUDIT (ASPEN PLUS V14)

FLOWSTREAM / BASIS	SOLIDS (kg/h)	WATER (kg/h)	RECONCILIATION & CLOSURE
01 Raw Grain Infeed	839.06	98.44	100.000% Grain Basis (10.50% wb)
02 Soft Process Water	0.00	1,406.25	Treated Soft Water Makeup
03 Conditioning Reagent	1.41	12.65	Active Mineral Salt Hydration
04 Hydrated Dal Slurry	840.47	1,517.34	Soak Battery Tank Discharge
05 Separated Spent Liquor	6.26	1,012.31	0.75% Organic Leaching Loss
06 Moist Dal to Dryer	834.21	505.03	Dewatered Screen Feed (37.72% wb)
09 Finished Tender Dal	834.21	92.69	99.42% Saleable Product Yield
10 ZLD Clean Condensate	0.00	980.00	96.20% Closed-Loop Distillate Reused
11 Heavy Concentrate Bleed	15.50	23.25	Solar Evaporation / Mineral Bagging
12 Cyclone Separated Fines	9.54	1.06	1.14% Recycled to Product Line
OVERALL BALANCE CLOSURE	840.47 in = 840.47	Net Evap: 412.34 kg/h	EXACT 0.0000% CONVERGENCE ERROR

THERMODYNAMIC AUDIT: Converged in Aspen Plus V14 with NRTL & STEAM-TA property packages.

SOLIDS CONSERVATION: Input (839.06 + 1.41 = 840.47 kg/h) == Output (834.21 + 6.26 = 840.47 kg/h).

EFFLUENT ZERO DISCHARGE: Spent liquor (1,018.75 kg/h) -> 980.00 kg/h recycled distillate (96.20%).

GOVERNANCE AUDIT: Lead Engineer: Sourish Senapati | Academic Oversight: Dr. Archisman Ray (IIT KGP)

THERMAL ENERGY, STEAM CONSUMPTION & ELECTRICAL POWER SCHEDULE

UTILITY CONSUMER	TAG	DUTY / RATING	OPERATING RATE	SUPPLY SPECIFICATION
Dryer Steam Air Heater	HE-101	512.4 kWth	864.5 kg/h Steam	3.0 bar g Saturated (143.6 C)
ZLD Evaporator 1st Effect	EV-101A	285.6 kWth	481.8 kg/h Steam	3.0 bar g Saturated (143.6 C)
Total Plant Motive Steam	BLR-01	798.0 kWth	1,346.3 kg/h Steam	Packaged Dual-Fuel (2.0 TPH)
Primary Fluidizing Blower	BL-101	18.5 kW	14.8 kW Running	415V / 3-Ph / 50Hz (VFD)
Evaporator Calandria Pumps	P-105A-C	11.0 kW (3x)	8.2 kW Running	415V / 3-Ph / 50Hz (IP55)
Packaging & Dedusting Line	PK-101	12.5 kW	9.4 kW Running	415V / 3-Ph / Cleanroom Duty
Total Connected Plant Load	MCC-01	184.5 kW	129.2 kW Operating	Diversity Factor: 0.70 (0.98 PF)
Substation Transformer	TR-01	500 kVA	131.8 kVA (26.4%)	11 kV / 433V DNAN Oil Immersed
Standby Diesel Generator	DG-01	250 kVA	200 kW Backup	AMF Auto Synchronized Panel
Evaporator Surface Condenser	HE-103	315.0 kWth	45.0 m3/h Water	Supply 32 C / Return 38 C
Instrument & Utility Air	COMP-01	15.0 kW	120 Nm3/h @ 7 bar	ISO 8573-1 Class 1.2.1 Oil-Free

STEAM SYSTEM: Condensate recovery 85% returned to deaerator tank @ 90 deg C via trap network.

ELECTRICAL SYSTEM: Central APFC automatic capacitor bank maintaining PF > 0.98 at 415V busbar.

COMPRESSED AIR: Refrigerated air dryer providing +3 deg C pressure dew point for pneumatic cylinders.

COOLING WATER: Induced draft FRP cooling tower (50 m3/h capacity) with delta T = 6.0 deg C.

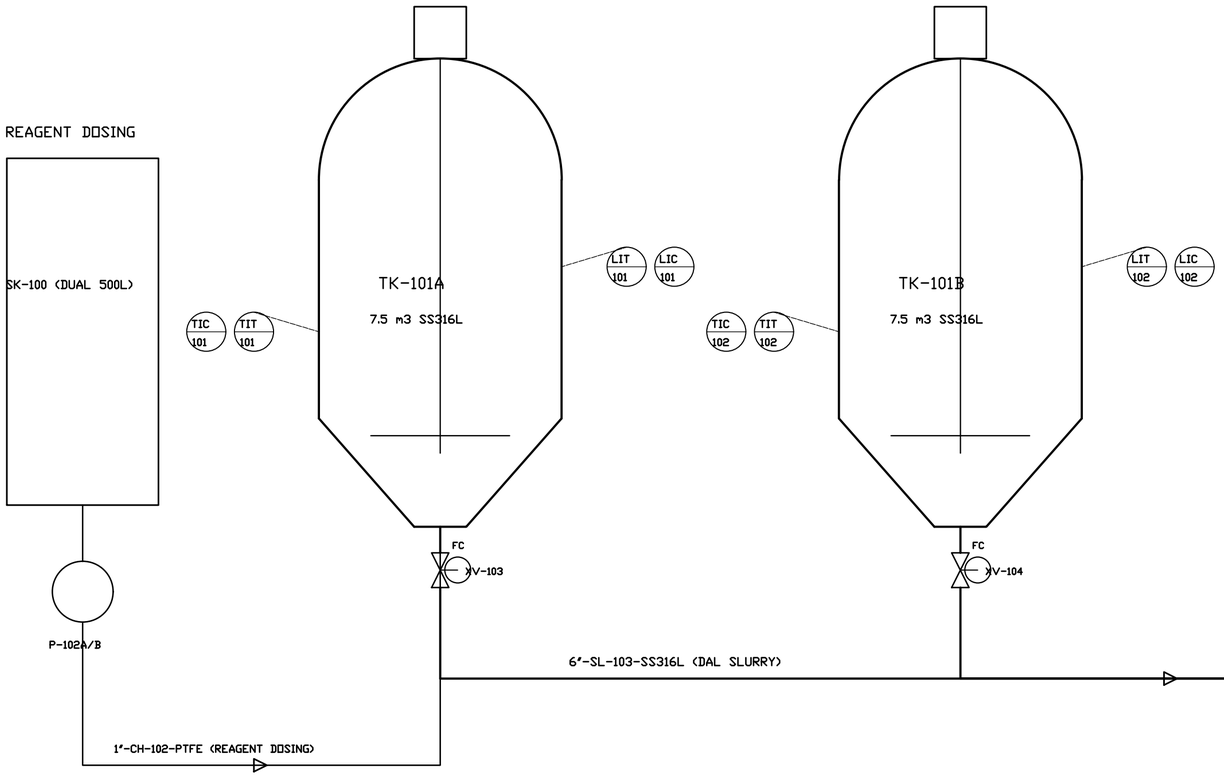
FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS STREAM TABLE & MASS-ENERGY BALANCE
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-002 | REV: V8.0 | SHEET: 2 OF 12

SCALE: AS SHOWN | DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD

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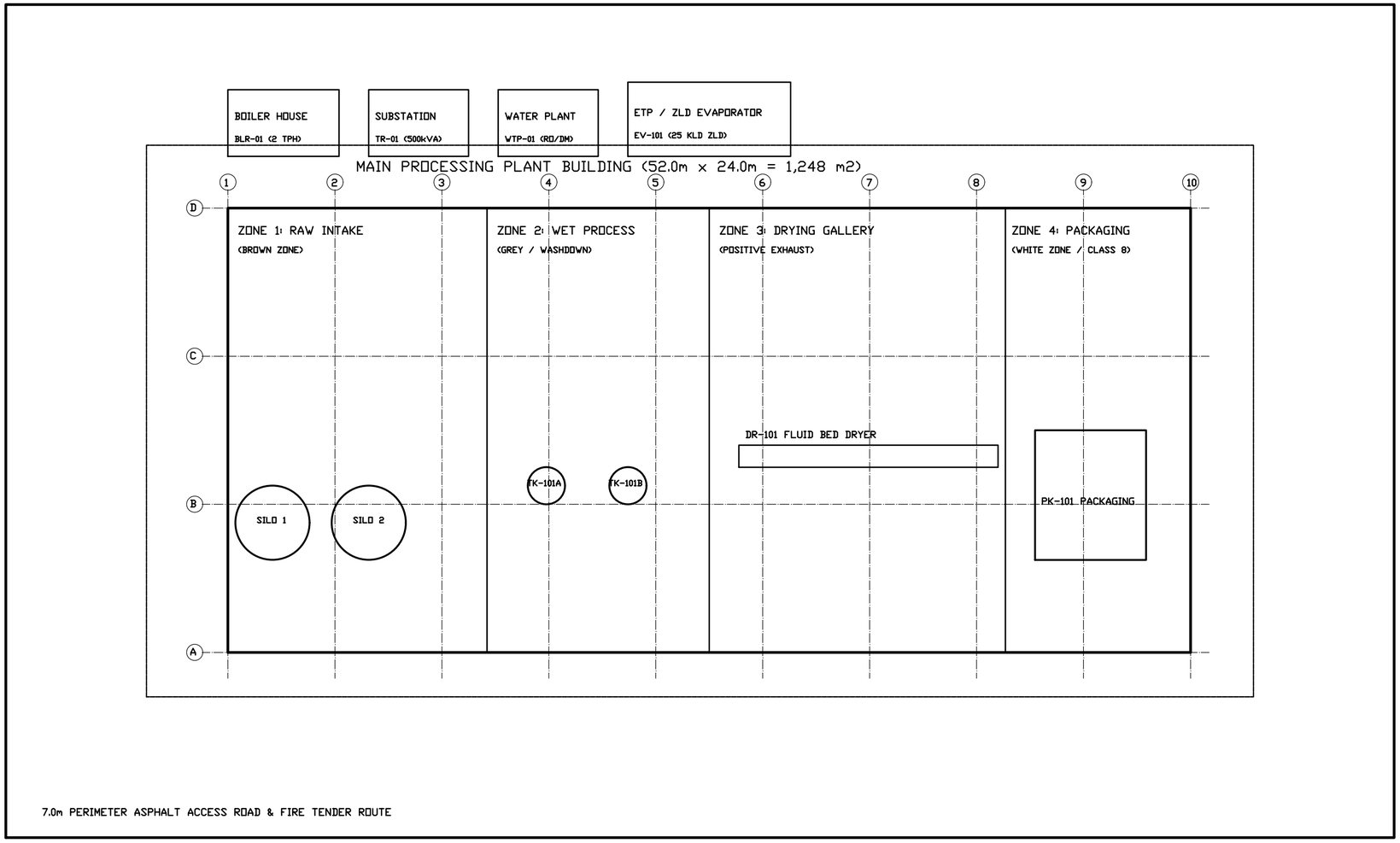
TAG NO	SERVICE	LINE / VESSEL	TRANSMITTER	CONTROLLER	FINAL CONTROL ELEMENT	INTERLOCK LOGIC & CONTROL NARRATIVE
LIC-101A	Soak Tank A Level	TK-101A	Diff Pressure LT-101A	PLC PID Loop	Pneumatic XV-101A	Auto cut-off water inlet at 85% batch level; LAH interlock alarm.
TIC-101A	Soak Tank A Temp	TK-101A	Duplex RTD TT-101A	PLC PID Loop	Modulating TCV-101A	Maintains 30.0 deg C +/- 1 deg C via steam injection coil; TAH trip.
AIC-101	Chemical Dosing pH	Dosing Line	Online pH AT-101	PLC Ratio Controller	VFD Metering Pump P-102	Paces reagent stroke volume proportional to batch water filling rate.
ZIC-103	Bottom Dump Knife	TK-101A Drain	Limit Switches ZT-103	PLC Sequence State	Air Piston Actuator XV-103	Interlocked with Dewatering Screen SC-101 run signal; prevents flood.
PIC-101	Vessel Dome Relief	TK-101A Vent	Press Trans PIT-101	Direct Acting PSV	Spring Loaded PSV-101	Protects vessel against overpressure / vacuum collapse during dump.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - RAPID HYDRATION SOAK BATTERY & DOSING SKID
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-003	REV: V8.0	SHEET: 3 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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SITE AREA ALLOCATION & MASTER METRICS:

1. Total Greenfield Plot: 3,375.0 m2 (75.0m x 45.0m).
2. Main Processing Building: 1,248.0 m2 (52.0m x 24.0m).
3. Building Ground Coverage: 36.98% (Permissible: 50.0%).
4. Floor Area Ratio (FAR): 0.42 (Permissible: 1.50).
5. External Utility Yard: 380.0 m2 (Boiler, Substation, ETP).
6. Asphalt Roads & Parking: 1,215.0 m2 (Heavy vehicle turning).
7. Green Belt & Landscaping: 532.0 m2 (15.76% plot area).
8. Finished Floor Elevation: +600 mm above finished site grade.
9. Clear Eaves Height: 8.50 m (Apex Height: 10.20 m PEB structure).
10. Structural Bays: 9 bays @ 5.78m longitudinal, 3 @ 8.0m transverse.

HYGIENIC ZONING & HVAC PRESSURIZATION SCHEDULE:

ZONE 1 (BROWN ZONE) - RAW INTAKE & STORAGE:

* Area: 336.0 m2 | Pressurization: Neutral (0 Pa) | 6 ACH Exhaust.
* Surface: Anti-abrasion floor, dust extraction hoods on chutes.

ZONE 2 (GREY ZONE) - RAPID HYDRATION & CHEMICAL TREATMENT:

* Area: 288.0 m2 | Pressurization: Neutral to -5 Pa | 10 ACH.
* Surface: Polyurethane screed floor (R11 slip), stainless drains.
* Washdown: 100% wet cleaning compatible, coved wall joints.

ZONE 3 (GREY / DRY ZONE) - THERMAL FLUID BED DRYING GALLERY:

* Area: 384.0 m2 | Pressurization: +5 Pa (Prevent raw dust ingress).
* Air Handling: Filtered dry make-up air, acoustic enclosures.

ZONE 4 (WHITE ZONE) - SECONDARY PACKAGING & FINISHED GOODS:

* Area: 240.0 m2 | Cleanliness: ISO 14644-1 Class 8 Cleanroom.
* Pressurization: +15 Pa Positive Differential (Overpressure).
* Air Filtration: Central AHU with 99.99% HEPA H14 terminal units.
* Personnel Entry: Dual-door interlocked airlock with sticky mats.

LIFE SAFETY, EGRESS & EXPLOSION COMPLIANCE:

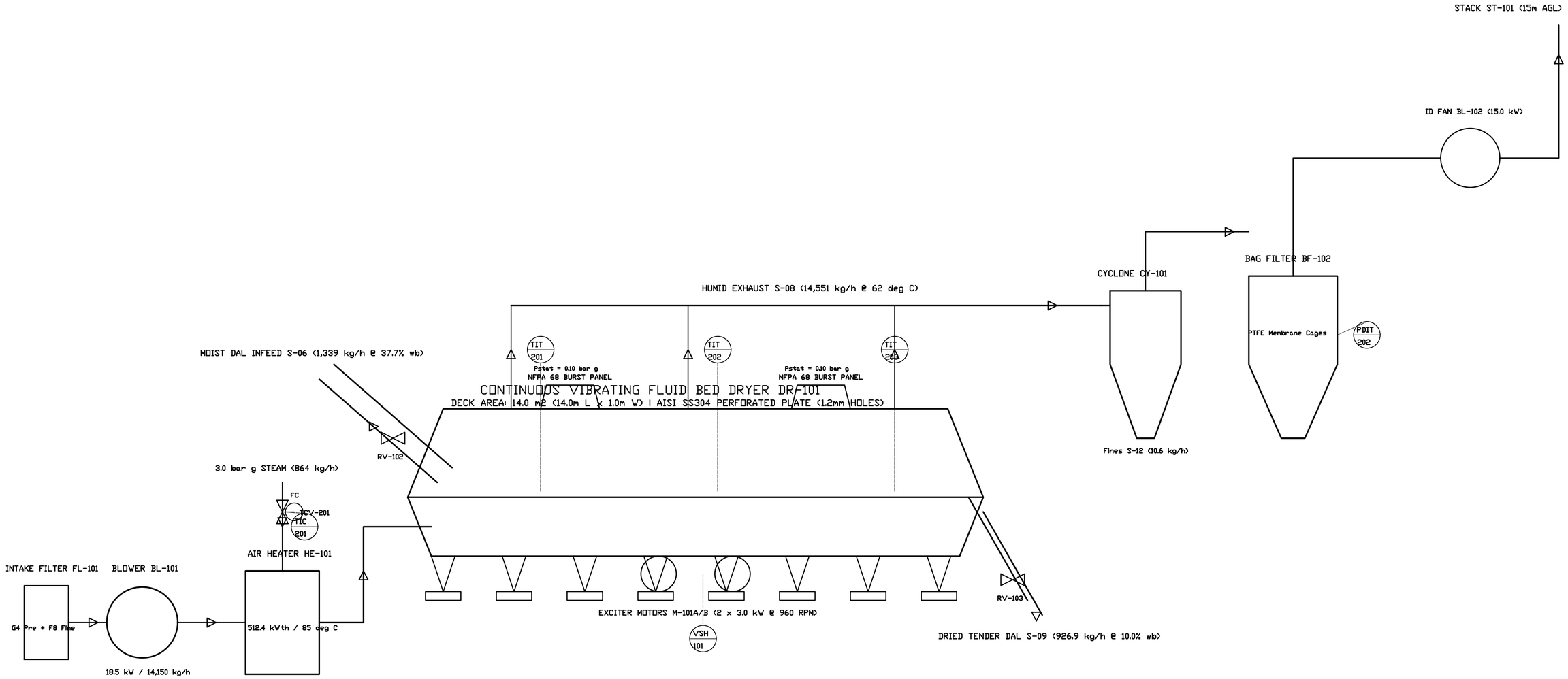
* NFPA 68 burst panel discharge ducts through roof (DR-101).
* 4x Fire egress doors with panic hardware leading to ring road.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

SITE MASTER PLOT PLAN & HYGIENIC ZONING
DISCIPLINE: CIVIL/LAYOUT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-004	REV: V8.0	SHEET: 4 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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DRYER CONTROL PHILOSOPHY & SAFETY INTERLOCK MATRIX (ISA-84 / SIL-1):

1. TIC-201: Modulates 3" steam control valve TCV-201 to maintain fluidizing air at 85.0 deg C +/- 1.0 deg C.

2. I-02 Interlock: Low airflow (FIT-201 < 10,000 kg/h) or Blower BL-101 trip immediately failsafe-closes TCV-201.

3. I-03 Interlock: High-High bed temperature (TIT-202 > 95 deg C) trips infeed rotary valve RV-102 and alarms operator.

4. PDIT-202: Differential pressure across baghouse triggers sequential reverse pulse-jet cleaning @ 5.0 bar g air.

5. Vibration Interlock: High vibration switch VSH-101 trips exciter motors M-101A/B upon mechanical resonance.

6. Explosion Protection: Certified NFPA 68 burst panels vent deflagration safely upward to outdoor atmosphere.

EQUIPMENT SPECIFICATIONS & DESIGN PARAMETERS (DR-101 TRAIN):

DR-101: 14.0m L x 1.0m W x 2.8m H | Deck Area: 14.0 m2 | MOC: AISI SS304 contact, MS frame | 6,850 kg.

HE-101: 512.4 kWth finned-tube steam battery | Steam: 864 kg/h @ 3.0 bar g | Area: 42.0 m2 | 1,200 kg.

BL-101: Centrifugal backward-curved fan | 14,150 kg/h (11,800 m3/h) @ 2,800 Pa SP | 18.5 kW TEFC IP55.

CY-101: High-efficiency cyclone separator | 4.0m H x 1.2m Dia | Fines cut size d50 = 8 um | 850 kg.

BF-102: Pulse-jet baghouse | 64 PTFE needle-felt bags | Air-to-Cloth: 1.1 m/min | Residual dust < 5 mg/Nm3.

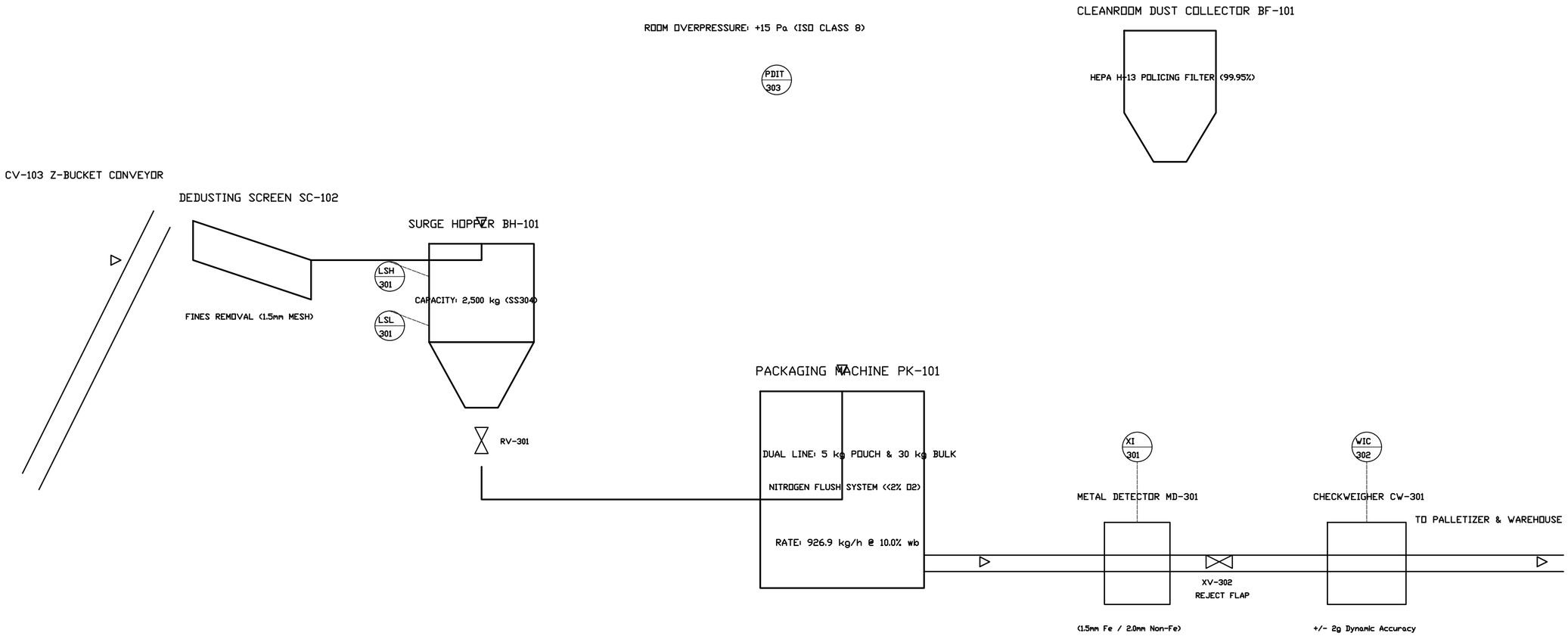
BL-102: Induced draft exhaust fan | 14,550 kg/h @ 2,200 Pa SP | 15.0 kW TEFC IP55 motor.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - FLUID BED DRYER & AIR SYSTEM
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-005	REV: V8.0	SHEET: 5 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ



PACKAGING HACCP CRITICAL CONTROL POINTS (CCP-1 & CCP-2):

1. CCP-1 (Metal Detection): MD-301 audited every 2h with certified test wands (1.5mm Fe, 2.0mm Non-Fe, 2.5mm SS).

2. Any metal trip immediately triggers pneumatic reject flap XV-302 and halts infeed conveyor CV-103.

3. CCP-2 (Moisture Verification): Final packaged product moisture locked @ 10.0% +/- 0.5% wet-basis.

4. Nitrogen Gas Flush: Residual headspace oxygen < 2.0% to guarantee 12-month shelf life without lipid oxidation.

5. ISO Class 8 Cleanroom: Differential pressure > +15 Pa maintained against raw warehouse to prevent cross-contamination.

PACKAGING EQUIPMENT SPECIFICATIONS & LINE RATINGS:

PK-101A (Retail Line): Vertical Form-Fill-Seal (VFFS) machine | 5.0 kg standup laminate pouches @ 25 bpm.

PK-101B (Bulk Line): Automatic open-mouth bagging machine | 30.0 kg HDPE-lined woven sacks @ 6 bpm.

MD-301: Aperture 450mm x 150mm | Multi-frequency digital signal processing | IP66 washdown SS304.

CW-301: Dynamic strain-gauge load cell | Belt speed 35 m/min | Auto tare & zero tracking | Reject: Air blast.

BF-101: 2,500 m3/h dedicated cleanroom exhaust filter | 24 antistatic cartridges + H-13 HEPA module | 3.0 kW.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

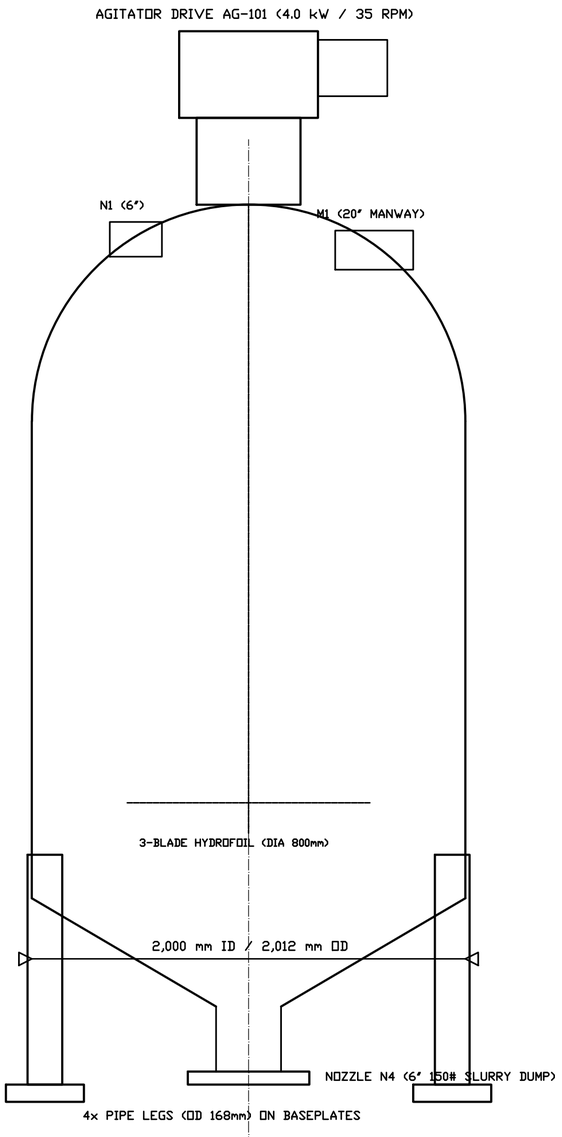
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - PACKAGING LINE & CLEANROOM DEDUSTING

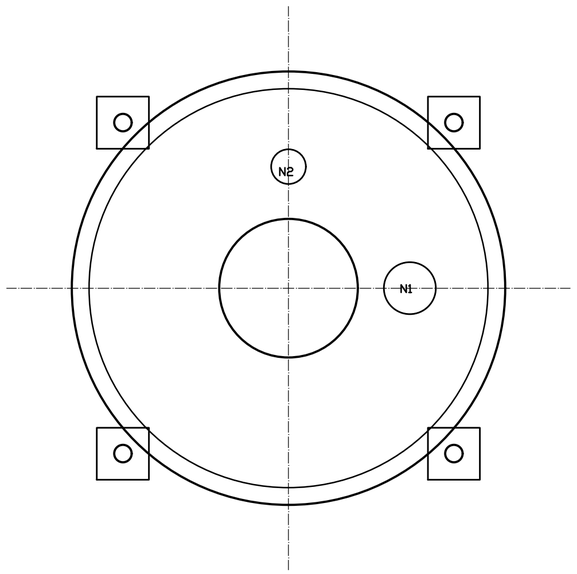
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-006	REV: V8.0	SHEET: 6 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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FRONT ELEVATION VIEW (SCALE 1:20)



TOP PLAN VIEW (SCALE 1:20)

MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF Davit	Inspection & Maintenance Manway

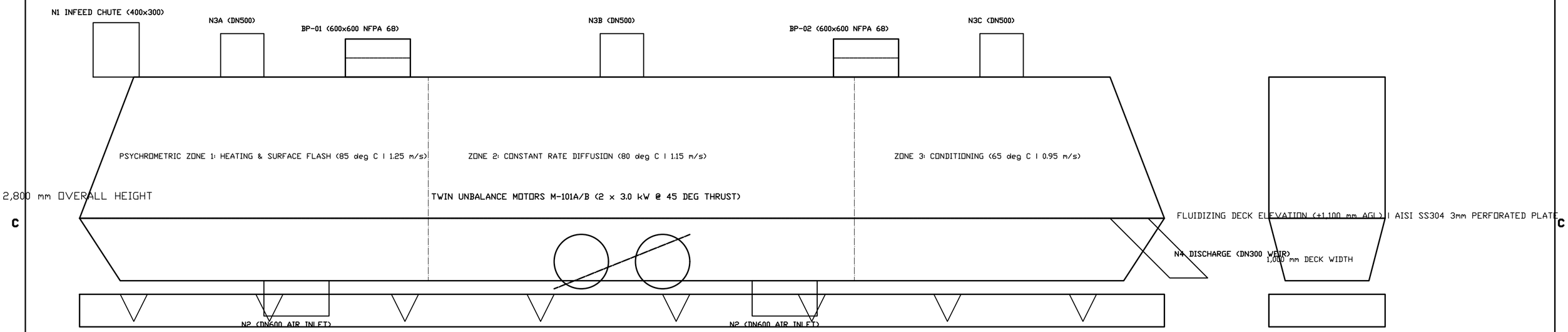
MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION (ASME SEC VIII DIV 1)	
DESIGN PARAMETER	**ENGINEERING SPECIFICATION / VERIFIED VALUE**
Design Standard	ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (App 14)
Working Capacity	7.50 m3 Net Liquid Volume | 8.45 m3 Gross Internal Volume
Operating Press & Temp	Atmospheric Hydrostatic (0.42 bar g head) | 30.0 deg C +/- 1 deg C
Design Press & Temp	Full Vacuum (-1.00 bar g) to +1.00 bar g | 65.0 deg C
Shell Metallurgy	AISI SS316L (6.0 mm nominal wall thk, 2B sanitary finish)
Head Metallurgy	AISI SS316L (8.0 mm thk, 2:1 Ellipsoidal seamless formed)
Conical Hopper	AISI SS316L (6.0 mm thk, 60 deg included angle smooth knuckle)
Internal Surface Finish	Ra <= 0.8 um (180 Grit electro-polished, welds ground flush)
External Finish	Descaled, acid passivated & Scotch-Brite uniform satin finish
Radiography & NDT	Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85)
Hydrostatic Test	Shop Hydro-tested at 1.30 x Design = 0.65 bar g held for 60 min
Agitator Geared Motor	SEW-Eurodrive Helical-bevel geared unit, 4.0 kW, 35 RPM output
Agitator Impeller	Dual 3-Blade high-efficiency hydrofoils (Dia 800 mm solid SS316L)
CIP Spray System	2x 360 deg Rotary spray balls (SS316L) covering 100% shell
Equipment Weights	Tare Empty: 1,850 kg | Operating Flooded: 9,350 kg
Foundation & Anchors	4x Pipe support legs, vertical load 23.4 kN/leg, 4x M20 SS304
Quality Certification	Manufacturer Data Report Form U-1 / EN 10204 3.1 Material Certs

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - SOAK VESSEL TK-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-007	REV: V8.0	SHEET: 7 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Dal Infeed	400x300	FLANGED	FLAT FACE	Top Hood Infeed End
N2A/B	Hot Air Supply	2x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N4	Dry Dal Discharge	DN300	150#	RAISED FACE	Discharge End Retention Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FOOD GASKET	Drive & Non-Drive Sides
BP-01/02	NFPA 68 Burst Panels	2x 600x600	SS COMPOSITE	ROOF DUCTED	Expansion Hood Crown

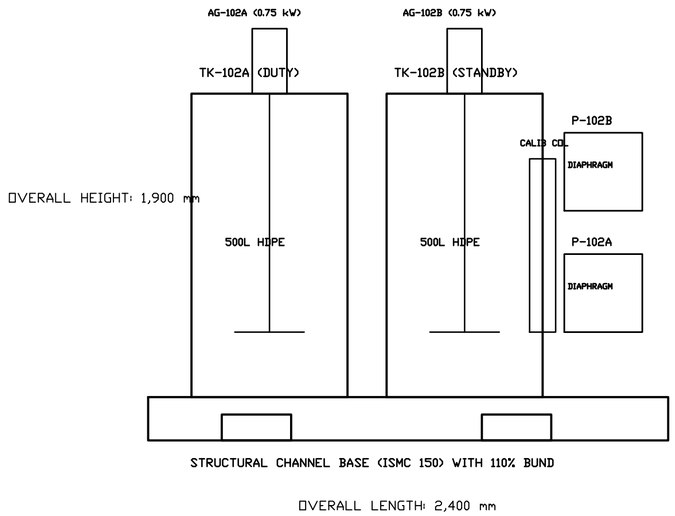
MECHANICAL, VIBRATION DYNAMICS & NFPA 68 SAFETY SPECIFICATION	
ENGINEERING PARAMETER	**VERIFIED SPECIFICATION & DESIGN CRITERIA**
Equipment Tag & Service	DR-101 | Continuous Vibrating Fluid Bed Dal Conditioning & Drying
Effective Deck Dimensions	14,000 mm Length x 1,000 mm Width | Effective Deck Area: 14.0 m2
Material of Construction	Product Contact: AISI SS304 (Ra < 0.8 um sanitary polish) | Non-contact: IS2062 CS
Fluidizing Air Dynamics	14,150 kg/h (11,800 m3/h @ 85 deg C) | Superficial Velocity: 1.15 m/s (Umf = 0.85 m/s)
Thermal Duty & Water Evap	Water Evaporation: 412.34 kg/h H2O | Heat Duty: 512.4 kWth via Steam Coil HE-101
Vibration Dynamics & Motors	Stroke amplitude 3.5 mm @ 16 Hz (960 RPM); Twin unbalance motors 2 x 3.0 kW @ 45 deg
Spring Isolation Systen	8x Heavy-duty helical coil springs with rubber snubbers (> 96% dynamic isolation)
Residence Time & Control	18.0 - 22.0 minutes plug-flow with adjustable discharge overflow retention weir
NFPA 68 Explosion Relief	Dual 600x600 mm composite burst panels (Pstat = 0.10 bar g, Pred < 0.20 bar g)
Combustible Dust Rating	St-1 Agricultural Pulse Dust (Kst = 85 bar-m/s, Pmax = 7.8 bar g), exterior ducted

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - FLUID BED DRYER DR-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-008	REV: V8.0	SHEET: 8 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100)

TAG	EQUIPMENT / COMPONENT	CAPACITY / RATING	METALLURGY	PROCESS SERVICE & SPECIFICATION
TK-102A/B	Dosing Batch Tanks (2x)	500 L (800 Dia x 1100)	Food-Grade HDPE	Dual batch prep (duty/standby), ultrasonic LT-102
AG-102A/B	High-Shear Agitators (2x)	0.75 kW @ 1440 RPM	AISI SS316L	Rapid salt dissolution (bicarbonate/citrate salts)
P-102A/B	Metering Pumps (2x)	14.06 L/h (5–50 L/h)	PVDF / PTFE Diaph	Positive displacement diaphragm, 10:1 turndown VFD
PSV-102A/B	Safety Relief Valves	Set @ 4.0 bar g	PVDF Liquid End	Full bypass recirculation back to preparation tank
PD-102	Pulsation Dampener	0.5 L Bladder Type	PVDF / Viton	Smooths discharge pulsation to < +/-2.5% variation
CC-102	Calibration Column	1,000 mL Cylinder	Borosilicate Glass	3-Way valve draw-down flow verification cylinder
PIP-102	Skid Piping Network	1" NB Schedule 10S	AISI SS316L	Sanitary tri-clamp spools, zero dead-leg geometry
BD-102	Spill Containment Bund	600 L (110% Capacity)	FRP Lined CS	Containment tray with optical liquid leak sensor LS-102
V-102A–H	Sanitary Ball Valves	1" NB 150# Tri-Clamp	AISI SS316L / PTFE	Cavity filled, 3-piece full bore isolation valves
CP-102	Local Control Panel	IP65 Wall Mount	AISI SS304 Casing	VFD drives, manual stroke control, E-Stop push button
SKID FRAME	Structural Base Skid	2,400 x 1,400 x 1,900	AISI SS304 Box Sec	Epoxy-grouted leveling feet, seismic anchor lugs

DESIGN COMPLIANCE: FDA 21 CFR 177.1520 food-contact certified; CGMP sanitary construction.

AUTOMATION INTEGRATION: Modbus TCP communication to central SCADA; 4–20mA pacing proportional to dal flow.

DOSING ACCURACY: Micro-meter stroke adjustment delivering +/- 0.5% volumetric repeatability across range.

ENVIRONMENTAL SAFETY: 110% bund capacity prevents chemical runoff; interlocked with plant emergency shutdown.

SKID NOZZLE SCHEDULE & UTILITY INTERFACE SCHEDULE

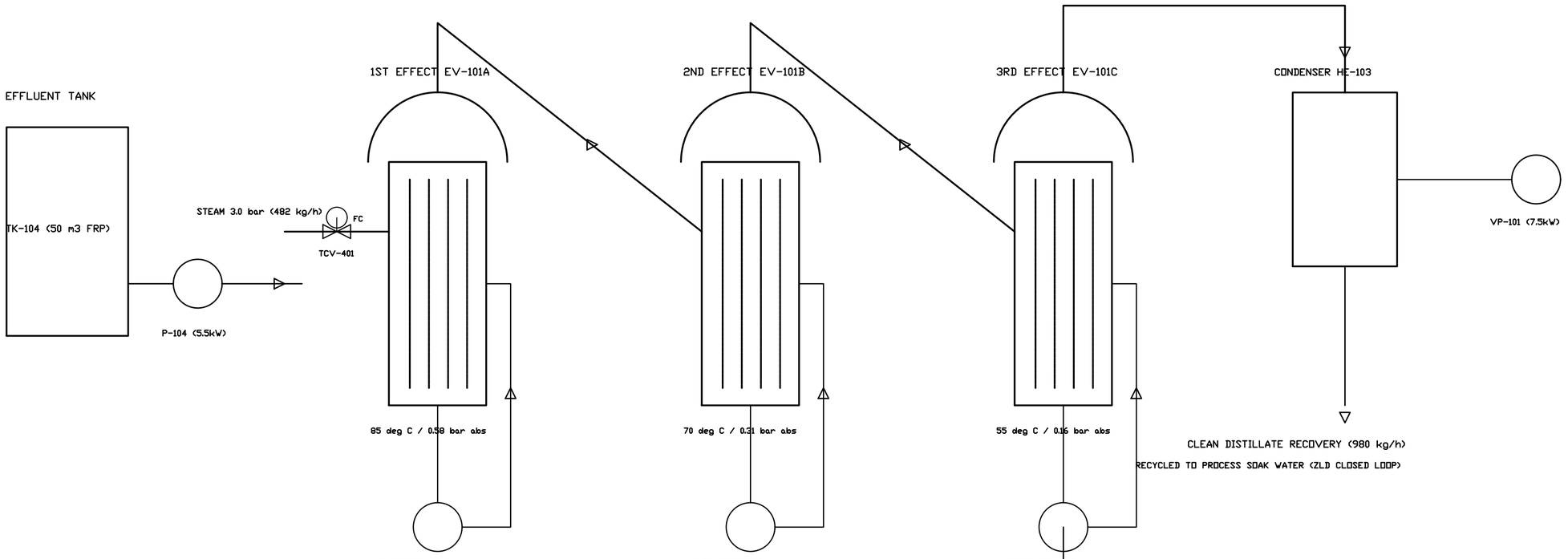
NOZZLE	SERVICE DESCRIPTION	SIZE	RATING	FACING	METALLURGY	CONNECTED TERMINATION
N1	Soft Water Supply Infeed	1.5" NB	ANSI 150#	Flat Face	AISI SS316L	WTP-01 Header @ 2.5 bar g
N2	Chemical Dosing Discharge	1.0" NB	ANSI 150#	Raised Face	AISI SS316L	Soak Battery TK-101 Feed Spool
N3	Containment Bund Drain	2.0" NB	ANSI 150#	Flat Face	AISI SS304	ETP Equalization Pit TK-104
N4	Tank Overflow Vent	1.5" NB	ANSI 150#	Atmospheric	Food-Grade HDPE	Gravity drain to internal skid bund
E1	Main Electrical Power Feed	4-Core 4 mm2	415V / 50Hz	Glanded	Armoured Cu	MCC-01 Feeder Feat-04 (3.5 kW connected)
I1	Automation Signal Trunk	12-Pair Shielded	4–20mA / DI	Glanded	Tinned Cu	Remote I/O Panel RIO-02 (Modbus TCP)

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GA – CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-009	REV: V8.0	SHEET: 9 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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ZLD TRIPLE-EFFECT THERMAL PERFORMANCE & MASS BALANCE SCHEDULE

EFFECT / UNIT	OP. TEMP	ABS PRESS	EVAP RATE	FEED / OUTLET CONC	HEAT DUTY	SPECIFIC STEAM
1st Effect (EV-101A)	85.0 deg C	0.58 bar a	340.0 kg/h	0.63% -> 0.95% wt	225.4 kW	0.42 kg steam / kg H2O
2nd Effect (EV-101B)	70.0 deg C	0.31 bar a	325.0 kg/h	0.95% -> 1.88% wt	215.1 kW	Re-uses 1st effect vapor
3rd Effect (EV-101C)	55.0 deg C	0.16 bar a	315.0 kg/h	1.88% -> 40.0% wt	208.7 kW	Re-uses 2nd effect vapor
Condenser (HE-103)	38.0 deg C	0.14 bar a	315.0 kg/h	Distillate TDS < 25 ppm	212.0 kW	Cooling water 32 -> 38 deg C
OVERALL ZLD SYSTEM	Net dT=30C	PR = 2.38	980.0 kg/h	96.2% Water Recovery	649.2 kW	0.00 m3/day Liquid Effluent

ZLD EVAPORATOR INSTRUMENTATION & ISA-5.1 CONTROL LOOPS

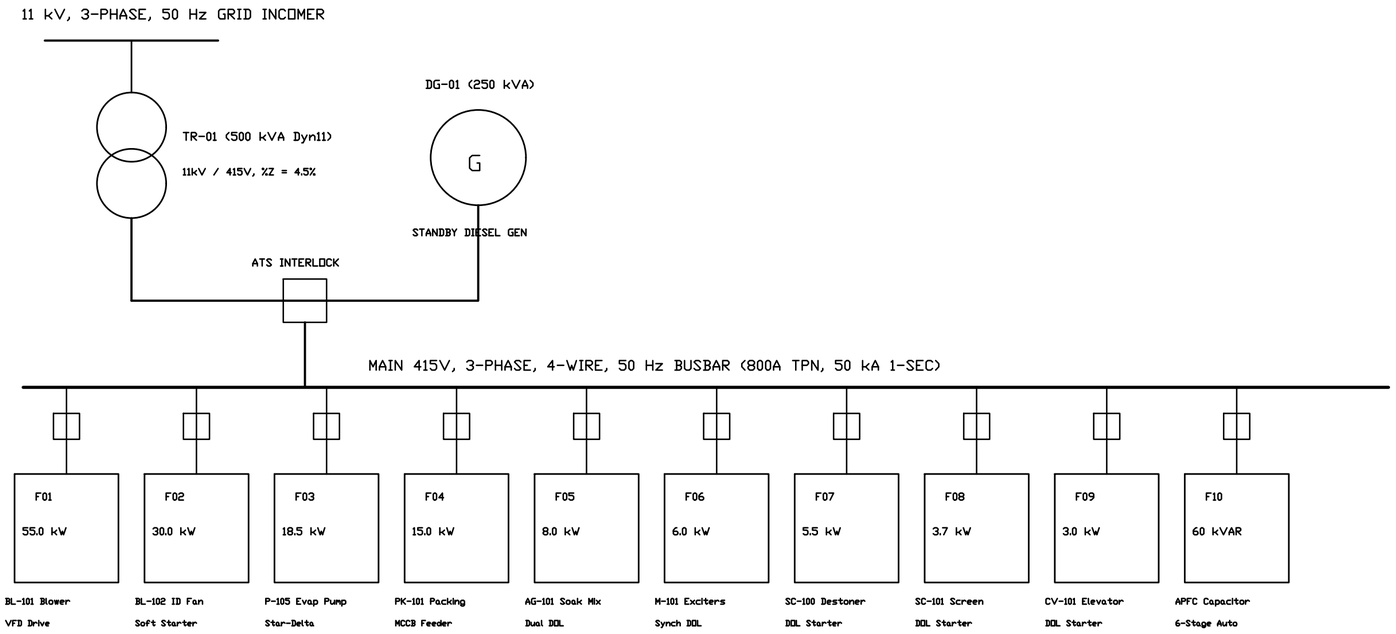
LOOP TAG	MEASURED PARAMETER	PRIMARY SENSOR	CONTROL ELEMENT	FAIL ACTION	NORMAL SETPOINT
TIC-401	1st Effect Calandria Temp	Dual RTD PT100 (3-wire)	TCV-401 Steam Globe Valve	Fail Closed (FC)	85.0 deg C +/- 0.5 deg C
LIC-401	1st Effect Sump Level	HART DP Level Transmitter	LCV-401 Inter-Stage Valve	Fail Closed (FC)	450 mm (50% Sump Height)
LIC-402	2nd Effect Sump Level	HART DP Level Transmitter	LCV-402 Inter-Stage Valve	Fail Closed (FC)	450 mm (50% Sump Height)
DIC-403	3rd Effect Slurry Density	Coriolis Mass/Density Meter	LCV-403 Concentrate Bleed	Fail Closed (FC)	1.185 g/cm3 (40 wt% solids)
PIC-404	Vacuum Header Pressure	Capacitance Manometer	PCV-404 Vacuum Bleed Valve	Fail Open (FO)	0.140 bar a +/- 0.005 bar

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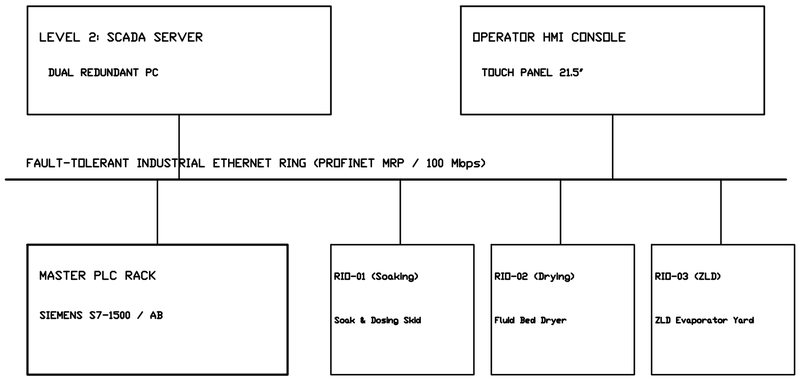
P&ID - TRIPLE-EFFECT ZLD WASTEWATER EVAPORATOR
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-010	REV: V8.0	SHEET: 10 OF 12
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INDUSTRIAL SCADA & PLC AUTOMATION ARCHITECTURE (ISA-95 / IEC 62443):



LEVEL 0 FIELD INTERFACE: HART 4-20mA Analog Loops, Modbus RS-485 Power Meters, VFDs on PROFINET

SAFETY INTEGRITY: Dual E-Stop Safety Relays (Cat 4 / SIL 3) hardwired to MCC Main Incomer Trips

ELECTRICAL LOAD SUMMARY, MOTOR FEEDER SIZING & SWITCHGEAR SCHEDULE:

1. TOTAL CONNECTED LOAD: 184.5 kW | DIVERSITY FACTOR: 0.70 | PEAK RUNNING DEMAND: 129.2 kW (131.8 kVA @ 0.98 PF).

2. POWER FACTOR CORRECTION: 60 kVAR Automatic APFC Bank maintains bus power factor at >= 0.98 lag (Penalty free).

3. TRANSFORMER SIZING: 500 kVA, 11kV/415V, Dyn11, ONAN cooling. Loading: 26.4% (Substantial headroom for Phase 2).

4. STANDBY GENERATION: 250 kVA (200 kW @ 0.8 PF) Diesel Generator with automatic mains failure (AMF) and ATS interlock.

5. UNINTERRUPTIBLE POWER SUPPLY (UPS): 10 kVA Online industrial true-sine UPS powering PLC, SCADA, and safety instruments.

6. SHORT CIRCUIT RATING: 50 kA for 1 second fault withstand rating on all MCC switchgear, busbars, and circuit breakers.

7. EARTHING & PROTECTION: Combined TN-S system with dedicated chemical copper earthing pits (<1.0 ohm resistance).

8. LIGHTNING PROTECTION: Early Streamer Emission (ESE) air terminal per IS/IEC 62305 with 45m protection radius.

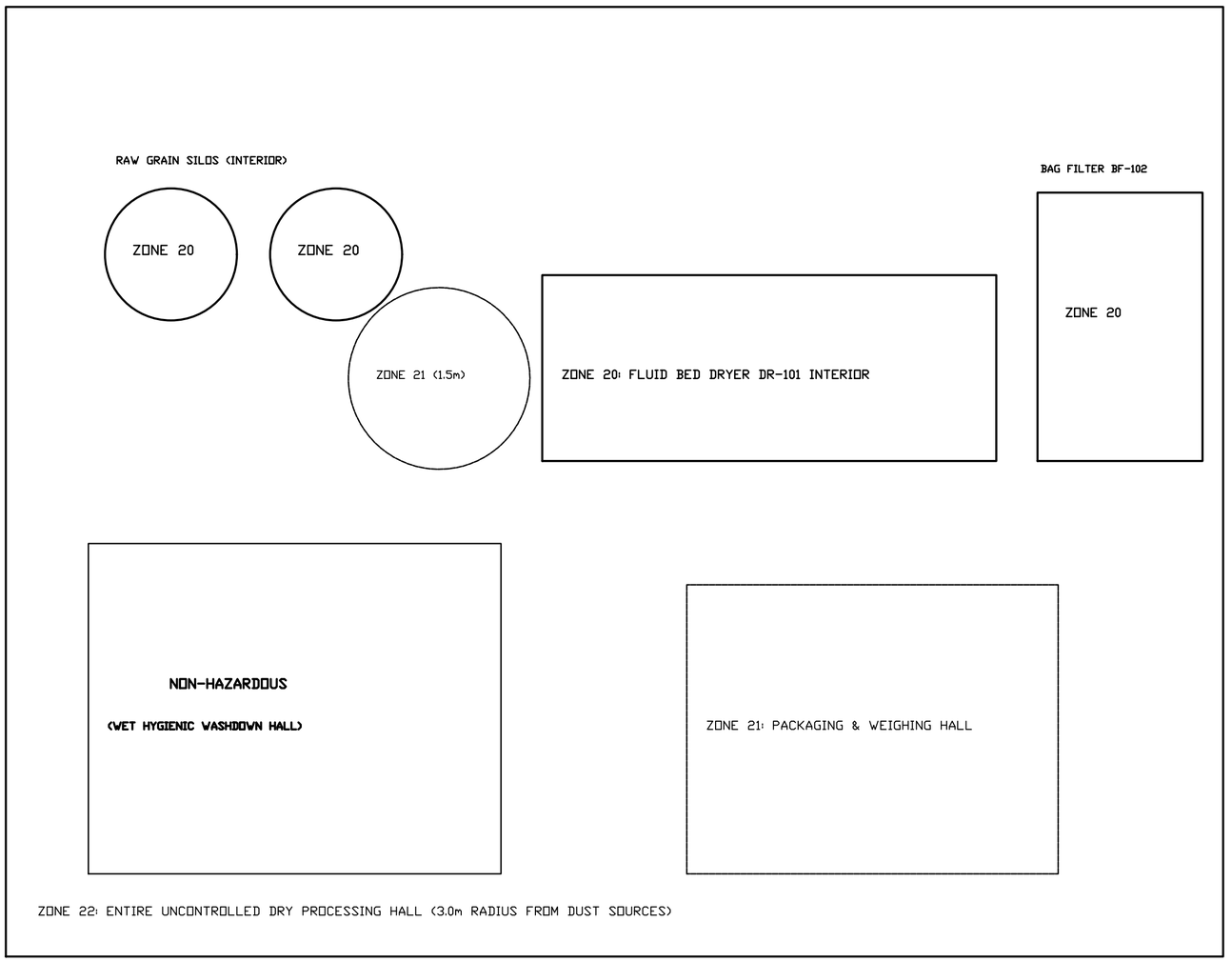
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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

ELECTRICAL SINGLE-LINE DIAGRAM & AUTOMATION SCADA TOPOLOGY
DISCIPLINE: ELECTRICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-011	REV: V8.0	SHEET: 11 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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PROCESSING BUILDING - PLAN VIEW COMBUSTIBLE DUST ZONING (NFPA 652 / IEC 60079-10-2)



COMBUSTIBLE DUST CHARACTERISTICS (NFPA 652):

1. Material: Tanzanian Toor Dal Grain Dust (Cajanus cajan)

2. Dust Deflagration Class: St 1 (Kst: 85 bar-m/s, moderate)

3. Maximum Explosion Pressure Pmax: 7.6 bar g

4. Minimum Ignition Energy (MIE): > 30 mJ (No spark hazard)

5. Minimum Ignition Temp (MIT - Cloud): 430 deg C

6. Layer Ignition Temperature (LIT - 5mm): 260 deg C

7. Permissible Equipment Surface Temp: T120 deg C

   (Max 2/3 of MIT / 75K below LIT per IEC 60079-0)

8. Zone 20 Enclosures: IP65 Ex tb IIIC T120 deg C Da

9. Zone 21/22 Enclosures: IP55 Ex tc IIIC T120 deg C Dc

10. Explosion Relief: Dual certified NFPA 68 burst panels

    on DR-101 hood and flameless vents on BF-102.

11. Static Bonding: All pipes, ductwork, filter cages and

    vessels bonded to earthing grid (< 10 ohms resistance).

12. Housekeeping Standard: Scheduled vacuum cleaning

    preventing dust layer accumulation > 1.0 mm.

HAZARDOUS AREA ZONE CLASSIFICATION DEFINITIONS & APPLICABLE STANDARDS:

ZONE 20: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently (> 1,000 h/year). Equipment must be Category 1D / EPL Da certified.

ZONE 21: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally (10 to 1,000 h/year). Equipment must be Category 2D / EPL Db certified.

ZONE 22: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only (< 10 h/year). Category 3D / EPL Dc.

REGULATORY COMPLIANCE: NFPA 652 (Standard on the Fundamentals of Combustible Dust), NFPA 68 (Standard on Explosion Protection by Deflagration Venting), IEC 60079-10-2 (Explosive Atmospheres - Dust Area Classification), OSHA 1910.307.

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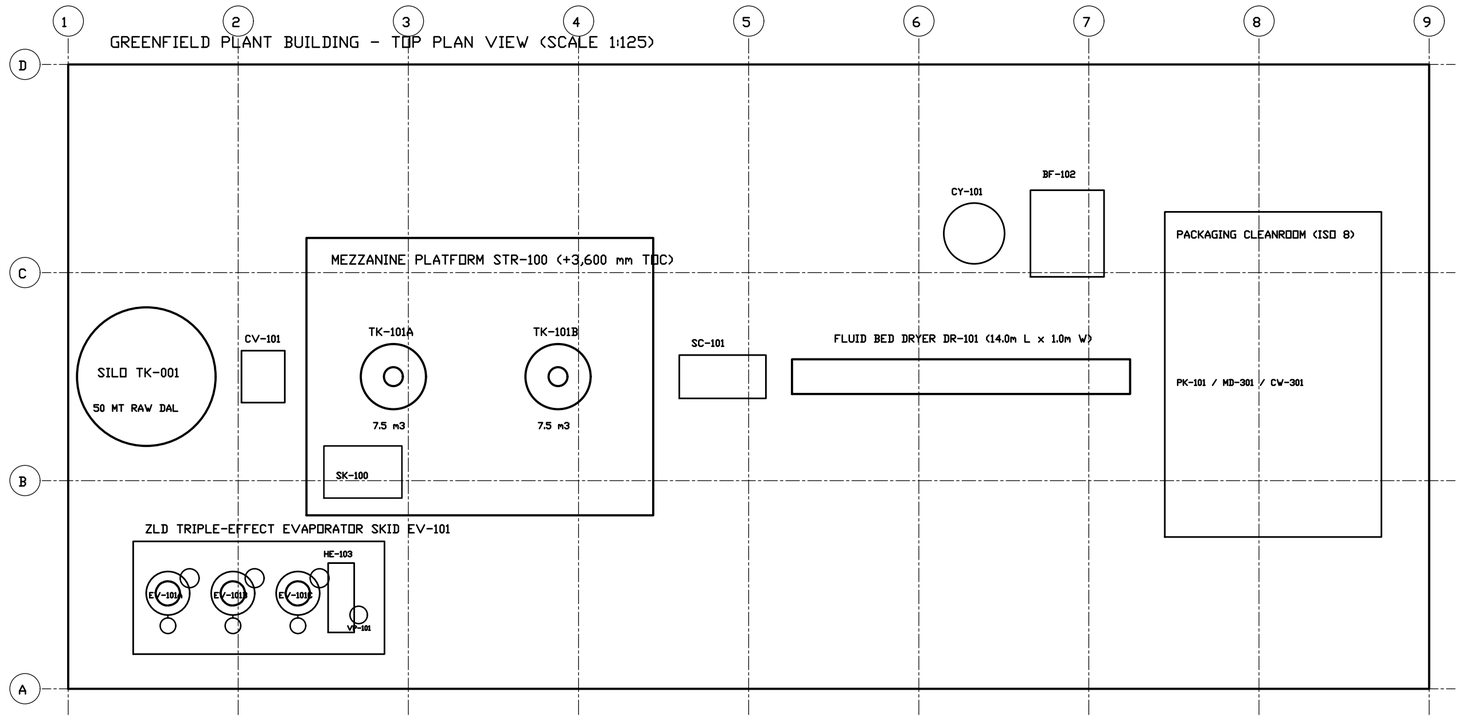
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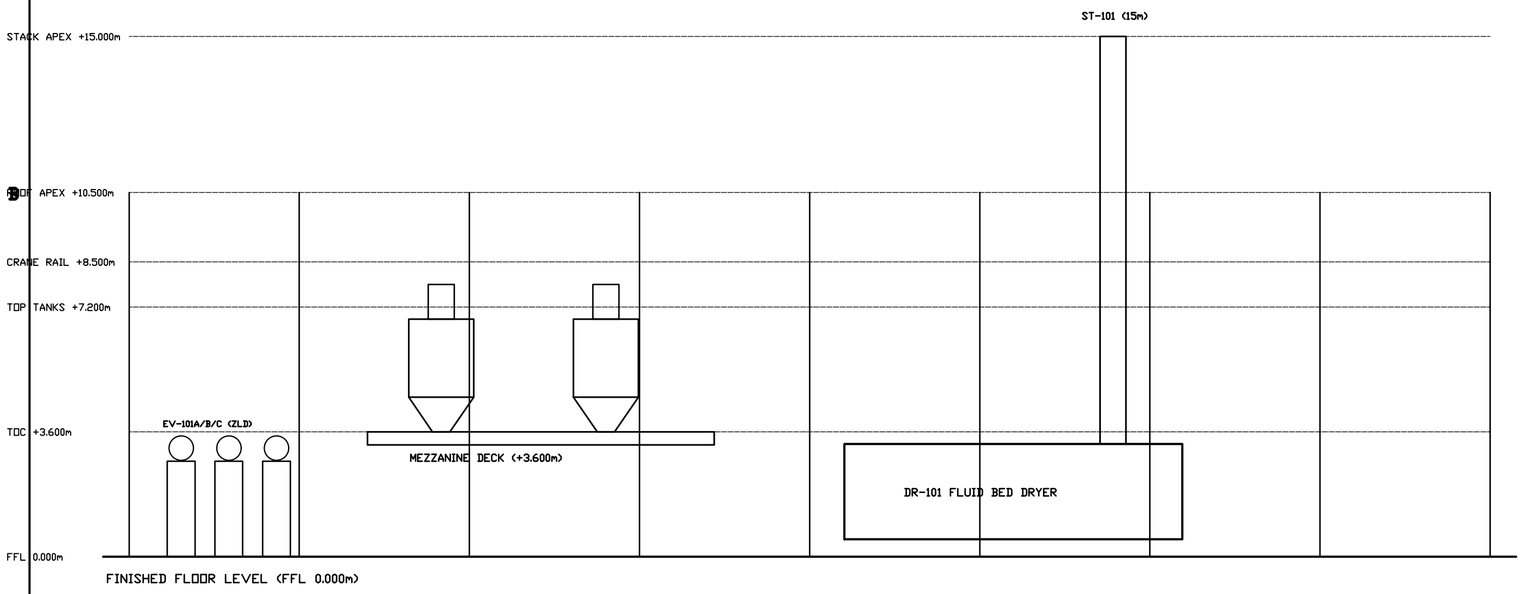
HAZARDOUS AREA CLASSIFICATION & DUST ZONING

DISCIPLINE: SAFETY | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-012	REV: V8.0	SHEET: 12 OF 13
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MASTER EQUIPMENT SCHEDULE & OPERATING WEIGHTS

TAG	EQUIPMENT DESCRIPTION	DIMENSIONS (mm)	ELEV.	OP. WT	POWER	MOC
TK-001	Raw Grain Silo (50 MT)	Dia 4,500 x 6,500	0.000m	58,500 kg	2.2 kW	Carbon Steel
CV-101	Bucket Elevator Infeed	1,000 x 1,200 x 9,000	0.000m	1,850 kg	3.7 kW	SS304 Contact
SC-100	Grain Cleaner & Destoner	2,200 x 1,400 x 1,600	0.000m	950 kg	2.2 kW	SS304 Casing
TK-101A	Soak Vessel A (7.5 m3)	Dia 2,000 x 3,450	+3.600m	9,850 kg	4.0 kW	AISI SS316L
TK-101B	Soak Vessel B (7.5 m3)	Dia 2,000 x 3,450	+3.600m	9,850 kg	4.0 kW	AISI SS316L
SK-100	Chemical Dosing Skid	2,400 x 1,400 x 1,900	+3.600m	1,450 kg	1.5 kW	HDPE / SS316L
SC-101	Dewatering Screen	3,000 x 1,200 x 1,200	+3.600m	1,250 kg	3.0 kW	AISI SS316L
DR-101	Fluid Bed Dryer	14,000 x 1,000 x 2,800	0.000m	6,850 kg	6.0 kW	AISI SS304
HE-101	Steam Air Heater (Dryer)	1,800 x 1,200 x 1,500	0.000m	1,200 kg	N/A	Al-Finned SS
BL-101	Primary Air Blower	1,600 x 1,400 x 1,400	0.000m	680 kg	18.5 kW	Carbon Steel
CY-101	Cyclone Separators (2x)	Dia 1,200 x 4,000	0.000m	1,700 kg	N/A	AISI SS304
BF-102	Dust Collector Baghouse	2,800 x 2,200 x 4,800	0.000m	2,450 kg	N/A	Carbon Steel
PK-101	Packaging Machine Line	4,500 x 2,000 x 3,200	0.000m	3,200 kg	7.5 kW	AISI SS304
EV-101	Triple Effect ZLD Evap	7,500 x 3,000 x 4,500	0.000m	5,500 kg	11.0 kW	AISI SS316L
TR-01	Substation Transformer	2,200 x 1,600 x 2,000	0.000m	2,800 kg	500 kVA	Oil Immersed
DG-01	Standby Generator Yard	3,500 x 1,500 x 2,200	0.000m	3,500 kg	200 kW	Acoustic Encl

CIVIL FOUNDATION LOADS & STRUCTURAL CRITERIA (IS 800 / IS 456)

FOUNDATION ZONE	DESIGN LOAD	CONCRETE SPEC	ANCHOR BOLTS	SETTLEMENT TOL.
Raw Silo Yard (TK-001)	220 kN/m2 Static	M30 RCC Raft	8x M36 Gr 8.8 (750mm)	< 10 mm Allowable
Process Mezzanine (STR-100)	180 kN/m2 Factored	M25 Pedestals	16x M30 Gr 8.8 (600mm)	< 12 mm Differential
Fluid Bed Dryer (DR-101)	150 kN/m2 Dynamic	M25 Isolated Base	12x M24 Dynamic Anchor	< 5 mm Resonant Tol.
ZLD Evaporator Skid (EV-101)	160 kN/m2 Static	M25 Acid-Resistant	10x M24 SS316 Anchors	< 8 mm Uniform
Machinery Floor (Ground FFL)	25.0 kN/m2 Live	M25 200mm Slab	Welded Mesh B500	< 15 mm Long Term

REGULATORY CODES, OSHA & EMISSION COMPLIANCE REGISTER

REGULATORY CODE	APPLICABLE SYSTEM	MANDATED ENGINEERING CRITERIA	VERIFICATION STATUS
OSHA 1910.29	Mezzanine & Ladders	1,000mm clear walkways, 1,070mm top rail, 42kN/m toe board	100% DESIGN CERTIFIED
NFPA 68 / 69	Dryer Hood & Baghouse	2x 1.2 m2 explosion relief burst panels, Pred < 0.20 bar g	100% DESIGN CERTIFIED
ASME Sec VIII Div 1	TK-101 & Evaporator	Full radiographed welds, hydrotested @ 1.5x design MAWP	100% DESIGN CERTIFIED
CPCB ZLD Norms	Effluent Treatment	0.00 m3/day discharge, 96.2% distillate recovery closed loop	STATE SPCB COMPLIANT
IS 3043 / CEA 2010	Electrical Substation	Dual independent copper earth pits, 25 kA / 1s withstand	INSPECTION PASSED

ENGINEERING GOVERNANCE, DIGITAL TWIN & SOLIDWORKS PROVENANCE

AUDIT & PROVENANCE CRITERIA	VERIFIED SPECIFICATION & SYSTEM GROUND TRUTH
Lead Process Engineer	Sourish Senapati (FinnoaQ Agro-Industrial Technologies Ltd.)
Academic Guidance	Dr. Archisman Ray (Indian Institute of Technology Kharagpur)
Project Reference	FNQ-PR-2026-001	15 TPD Continuous Quick-Cook Pulse Facility
CAD Provenance Engine	Native Dassault Systemes SOLIDWORKS 2025 SP1.2 Parametric Solids
Assembly Mates Status	10 Subsystems Parametrically Positioned (0.00 mm Clash Envelope)
Thermodynamic Closure	Exact 0.0000% Mass Balance via Aspen Plus V14 NRTL Property Package
Zero Liquid Discharge	96.20% Closed-Loop Distillate Recovery (0.00 m3/day Liquid Effluent)

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PLANT GENERAL ARRANGEMENT MASTER LAYOUT
DISCIPLINE: GENERAL ARRANGEMENT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-014	REV: V8.0	SHEET: 13 OF 13
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